

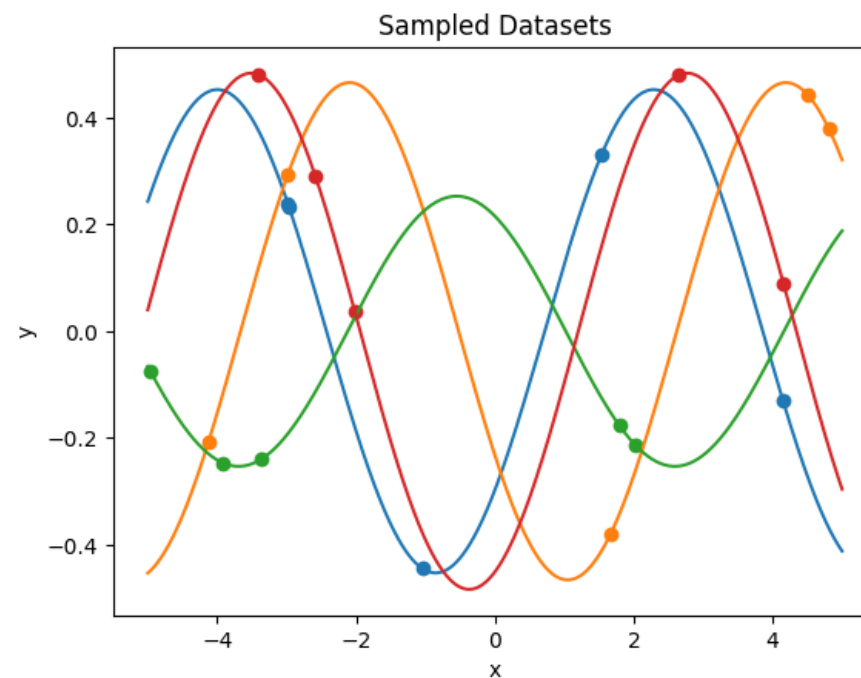
Exercise: Meta Learning for Static Regression

The “sine” meta learning example

- Sines of varying phase and amplitude

$$y(x) = A \sin(x + \psi)$$

- Simple dependency, but just $K = 5$ points per dataset!
- Taken from the famous MAML paper
- The “Hello World” Meta Learning problem!



Meta Learning Setting

- We have access to a *meta dataset*, namely a collection (possibly unlimited) of datasets

$$\mathcal{D} = \{D_1, D_2, \dots, \}$$

- In the case of static regression, each dataset D_i is an unordered collection of K input-output pairs.

$$D_i = \{(x_{i,1}, y_{i,1}), (x_{i,2}, y_{i,2}), \dots, (x_{i,K}, y_{i,K})\}, \quad x_{i,j} \in \mathbb{R}^{n_x}, \quad y_{i,j} \in \mathbb{R}^{n_y}$$

- The datasets D_i are assumed to be *similar* to each other. They are thought as realizations from a probability distribution $p(D)$
- Each dataset D_i is split in a train and a test portion:

$$\mathcal{D} = \{(D_1^{\text{tr}}, D_1^{\text{te}}), (D_2^{\text{tr}}, D_2^{\text{te}}), \dots\}$$

Meta Learning: In-context learning

- We define an **meta-model** $\hat{y} = \mathcal{M}_\phi(x, D)$ which takes as input:
 - a test point x
 - a training dataset D (i.e., K input/output data points).It produces as output the prediction $\hat{y}$ corresponding to input x , given as a *context* the dataset D .
- Implicitly, the meta-model estimates a *system-specific* model on D and applies it to x
- We train the meta model by minimizing over the meta-dataset:

$$J(\phi) = \sum_{i=1}^b \sum_{j=1}^K \mathcal{L}(y_{i,j}^{\text{te}}, \mathcal{M}_\phi(D_i^{\text{tr}}, x_{i,j}^{\text{te}})).$$

- The meta model learns to predict the output with just K input/output examples

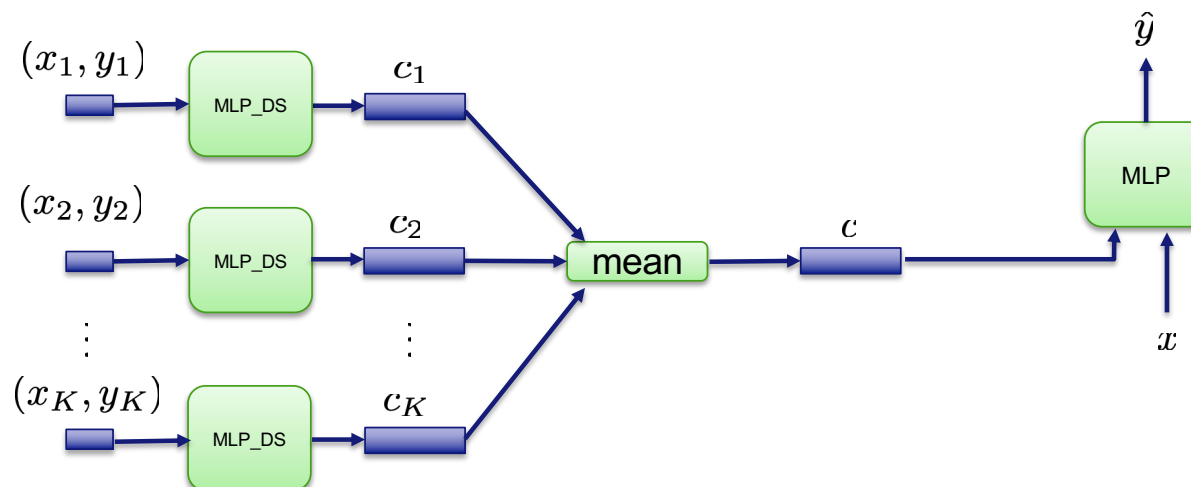
In-context learning architecture for static regression

- The meta-model $\hat{y} = \mathcal{M}_\phi(D, x)$ has structure:

$$c = \text{DeepSets}_{\phi_1}(D)$$

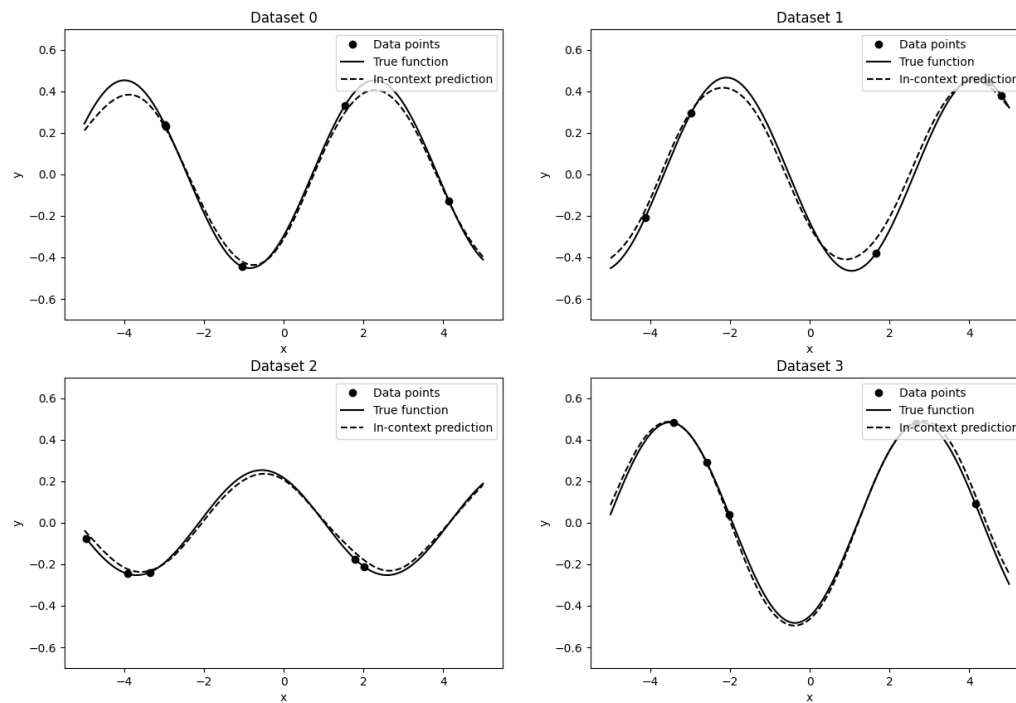
$$\hat{y} = \text{MLP}_{\phi_2}(c, x).$$

DeepSets is *invariant to permutations* of the K input/output examples in D



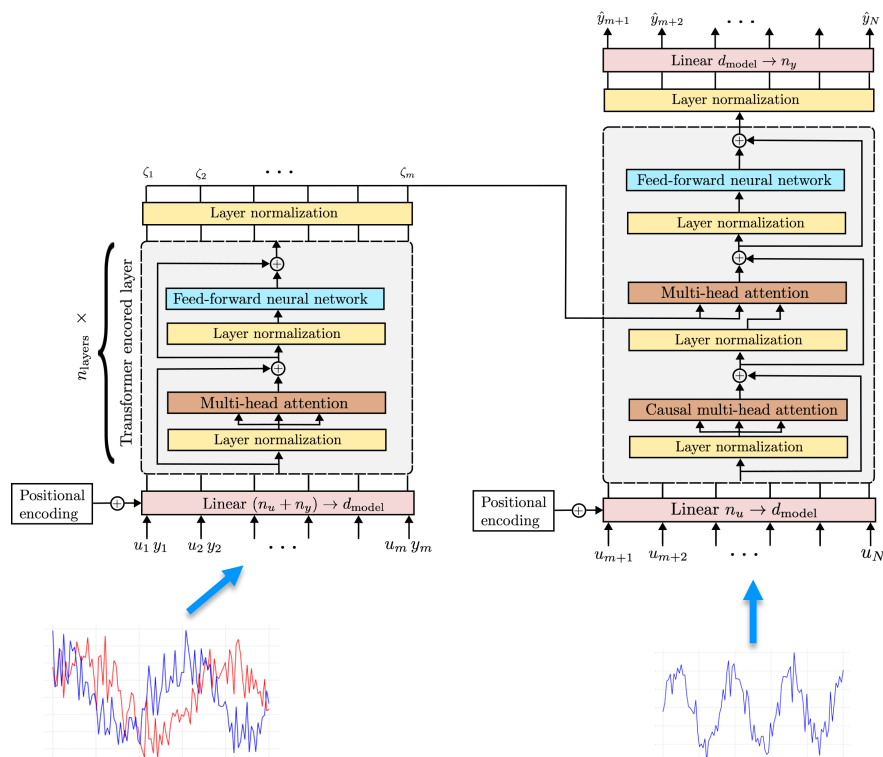
Performance with meta learning (in-context learning)

Sampled datasets



The meta model learns to make pretty good predictions with $K = 5$ data points

In-context learning architecture for system identification



- May be seen as a simplification of the encoder-decoder Transformer for system identification

$$\hat{y}_{m+1:N} = \mathcal{M}_{\phi}(u_{m+1:N}, u_{1:m}, y_{1:m})$$

- The Transformer's attention layers are ideal to deal with intricate temporal dependencies of SYSID.
- For the sines example, the simple DeepSet+MLP architecture is more than enough...

Exercise

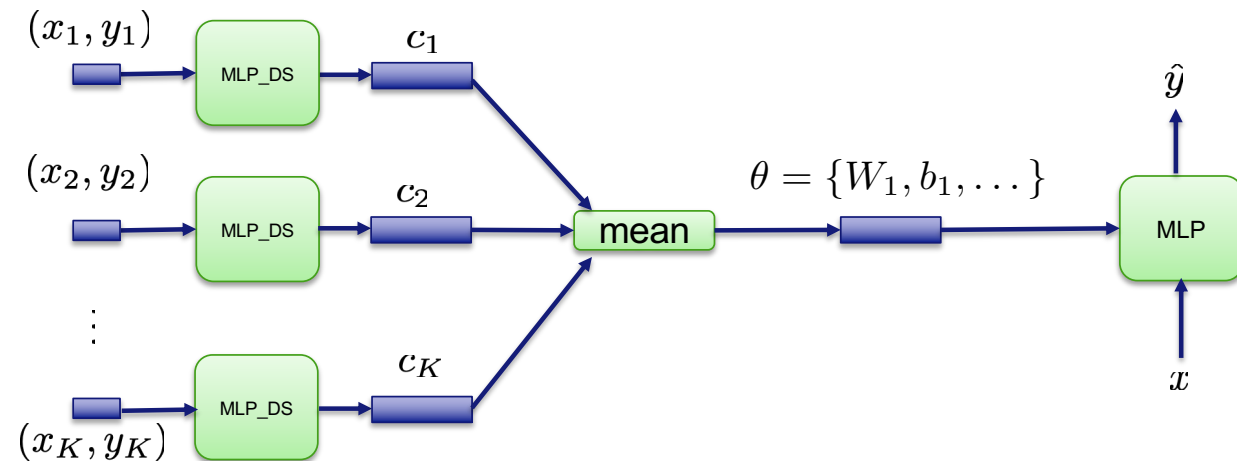
- Implementation of the in-context learning architecture for static regression DeepSets + MLP is sketched in `in_context_learning_sketch.ipynb`
- Use it to tackle the meta learning exercise (`meta_learning_exercise.ipynb`)
- Optionally, apply hyper-networks and MAML (sketched in `maml_hypernet_sketch.ipynb`, described in the next slides...)

Meta Learning with Hyper-Networks

- A hyper-network provides weight and biases of the model MLP describing the relation $x \rightarrow y$

$$\theta = \text{DeepSets}_{\phi}(D)$$

$$\hat{y} = \text{MLP}_{\theta}(x).$$



- The hyper-network may be seen as a *learned* system identification algorithm

Gradient-Based Meta Learning (MAML)

- The learned algorithm has much more structure: it is one step of gradient descent!

$$\begin{aligned}\theta &= \text{Alg}_\theta(D) = \phi - \alpha \nabla_\theta \mathcal{L}(\theta, D), \\ \hat{y} &= \text{MLP}_\phi(x).\end{aligned}$$

- We learn the best parameter ϕ such that, with one (or more) steps of gradient descent on the training dataset, performance measured on the test datasets is good.

$$J(\phi) = \sum_{i=1}^b \mathcal{L}(\text{Alg}_\phi(D_i^{\text{tr}}), D_i^{\text{te}})$$

Finn, Chelsea, Pieter Abbeel, and Sergey Levine. "Model-agnostic meta-learning for fast adaptation of deep networks." International conference on machine learning. PMLR, 2017.